Geoffrey X. Yu

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Research Interests Databases, Systems, Cloud-Native Databases, Workload-Adaptive Data Systems

Education

Massachusetts Institute of Technology 2020 – 2026
Doctor of Philosophy in Computer Science (PhD) Cambridge, MA

- *Advisor:* Tim Kraska, Data Systems Group
- *Dissertation:* Virtualizing Data Infrastructures

University of Toronto 2018 – 2020
Master of Science in Computer Science (MSc) Toronto, ON, Canada

- *Advisor:* Gennady Pekhimenko
- *Research:* GPU performance prediction for deep neural network training

University of Waterloo 2013 – 2018
Bachelor of Software Engineering (BSE) Waterloo, ON, Canada

Academic Appointments

Postdoctoral Associate September 2026 – Present
Massachusetts Institute of Technology Cambridge, MA

- *Host:* Tim Kraska, Computer Science and Artificial Intelligence Laboratory (CSAIL).

Publications

EXPLAIN Yourself! Finding Query Planner Stalls Across DBMSes
Geoffrey X. Yu, Ryan Marcus, Tim Kraska
Conference on Innovative Data Systems Research (CIDR), 2027. To appear. Preprint: arXiv:2608.23402.

Blueprinting the Cloud: Unifying and Automatically Optimizing Cloud Data Infrastructures with BRAD
Geoffrey X. Yu, Ziniu Wu, Ferdi Kossmann, Tianyu Li, Markos Markakis, Amadou Ngom, Samuel Madden, Tim Kraska
Proceedings of the VLDB Endowment (VLDB), Vol. 17, No. 11., 2024.

Check Out the Big Brain on BRAD: Simplifying Cloud Data Processing with Learned Automated Data Meshes
Tim Kraska*, Tianyu Li*, Samuel Madden*, Markos Markakis*, Amadou Ngom*, Ziniu Wu*, Geoffrey X. Yu*
Proceedings of the VLDB Endowment (VLDB), Vol. 16, No. 11, 2023. Vision Paper.

TreeLine: An Update-In-Place Key-Value Store for Modern Storage
Geoffrey X. Yu*, Markos Markakis*, Andreas Kipf*, Per-Åke Larson, Umar Farooq Minhas, Tim Kraska
Proceedings of the VLDB Endowment (VLDB), Vol. 16, No. 1, 2022.

Habitat: A Runtime-Based Computational Performance Predictor for Deep Neural Network Training
Geoffrey X. Yu, Yubo Gao, Pavel Golikov, Gennady Pekhimenko
USENIX Annual Technical Conference (USENIX ATC), 2021.

Skyline: Interactive In-Editor Computational Performance Profiling for Deep Neural Network Training
Geoffrey X. Yu, Tovi Grossman, Gennady Pekhimenko
ACM Symposium on User Interface Software and Technology (UIST), 2020.

* Denotes equal contribution.

Preprints and Under Submission

Tailwind: A Practical Framework for Query Accelerators
Geoffrey X. Yu, Ryan Marcus, Tim Kraska
arXiv Preprint arXiv:2604.28079, 2026.

ExperienceIndex: Artifact-Grounded Memory
Peter Baile Chen, Geoffrey X. Yu, Xinming Liu, Samuel Madden, Dan Roth, Doug Downey, Michael Cafarella

Under submission to the International Conference on Learning Representations (ICLR), 2027.
arXiv Preprint arXiv:2610.10091, 2027.

JetStream: Generating Verified Query Accelerators for Existing Database Systems
Akhilesh Balasingam, Amadou Ngom, Geoffrey X. Yu, Tim Kraska
arXiv Preprint arXiv:2610.20527, 2026.

Correctness Firewalls for Synthesized Database Components
Mohammad Javad Amiri, Carsten Binnig, Andreas Kipf, Tim Kraska, Ryan Marcus, Johannes Wehrstein, Chenyuan Wu, Geoffrey X. Yu, Zijie Zhao
Under submission to Proceedings of the VLDB Endowment (VLDB), 2026.
arXiv Preprint arXiv:2611.05173, 2026.

Demonstrations

Virtualizing Cloud Data Infrastructures with BRAD
Geoffrey X. Yu, Ziniu Wu, Ferdi Kossmann, Tianyu Li, Markos Markakis, Amadou Ngom, Sophie Zhang, Tim Kraska, Samuel Madden
International Conference on Management of Data (SIGMOD), 2025.

TBD Suite: Benchmarking and Profiling Tools for DNNs
Geoffrey X. Yu, Hongyu Zhu, Anand Jayarajan, Bojian Zheng, Abhishek Tiwari, Gennady Pekhimenko
Conference on Machine Learning and Systems (MLSys), 2019.

Professional Experience

Amazon Web Services — Applied Scientist Intern
Boston, MA — May – August 2024

- *Mentor:* Vikram Nathan.
- Led research on designing and simulating strategies for per-query configurable per-query configurable cost-performance trade-offs in Redshift Serverless' intelligent scaling mode.

Intel Corporation — Research Graduate Intern
Remote — May – August 2021

- *Mentors:* Nesime Tatbul and David Cohen.
- Led research on new designs for concurrent learned indexes that support database transactions.

Meta Platforms (formerly Facebook) — Software Engineering Intern
Menlo Park, CA — January – April & September – December 2017

Apple — Software Engineering Intern
Sunnyvale, CA — May – August 2016

- Contributed to Apache Cassandra to improve observability and data repair capabilities.

Coursera — Software Engineering Intern
Mountain View, CA — January – April & September – December 2015

Teaching Experience

Teaching Assistant — Massachusetts Institute of Technology
6.5830 / 6.5831 – Database Systems — Fall 2022

- *Instructors:* Samuel Madden and Michael Cafarella
- *Overall Rating:* 6.7 out of 7 ($\sigma = 0.48$, $n = 10$)

Teaching Assistant — University of Toronto
CSC 209 – Software Tools and Systems Programming — Fall 2018, Winter 2019

Student Mentoring

- **Sophie Zhang** — Massachusetts Institute of Technology
  Undergraduate research, Master's thesis (MEng) — 2024 – 2025
  ↪ Next: Software Engineer at Databricks

- **Yubo Gao** — University of Toronto
  Undergraduate research — 2019 – 2020
  ↪ Next: PhD student at the University of Toronto
- **Pavel Golikov** — University of Toronto
  Undergraduate research — 2019 – 2020
  ↪ Next: PhD student at the University of Toronto

Service

- VLDB 2026 Demonstration Track Program Committee
- EDBT 2027 Demonstration Track Program Committee
- VLDBJ Reviewer

Awards and Honors

- **NSERC Postgraduate Scholarship – Doctoral (PGS D)** — 2020 – 2023
  Merit-based doctoral research scholarship awarded by the Government of Canada (63,000 CAD).
- **NSERC Alexander Graham Bell Canada Graduate Scholarship (CGS D)** — 2020 – 2023 (declined)
  Merit-based doctoral research scholarship awarded by the Government of Canada (105,000 CAD).
- **NSERC Canada Graduate Scholarship – Master's (CGS M)** — 2019 – 2020
  Merit-based master's research scholarship awarded by the Government of Canada (17,500 CAD).
- **Snap Research Scholarship** — January 2019
  Awarded by Snap Inc. for research quality, technical skills, and communication skills (10,000 USD).
- **Vector Institute Scholarship in Artificial Intelligence** — 2018 – 2019
  Awarded to top students pursuing AI-related master's studies in Ontario (17,500 CAD).
- **Queen Elizabeth II Graduate Scholarship in Science and Technology** — 2018 – 2019
  Merit-based scholarship awarded to students pursuing graduate studies in Ontario (15,000 CAD).
- **Wolfond Fellowship, University of Toronto** — 2018 – 2019
  Awarded at the University of Toronto for academic achievements (10,000 CAD).
- **Sandford Fleming Foundation Award for Academic Excellence** — June 2018
  Best graduating academic record in the Software Engineering program at the University of Waterloo.

Invited Talks

- **Virtualizing Data Infrastructures** — November 3, 2026
  University of Wisconsin–Madison — Madison, WI
- **Tailwind: A Practical Framework for Query Accelerators** — May 21, 2026
  InterSystems Corporation — Boston, MA
- **Virtualizing Cloud Data Infrastructures with BRAD** — May 23, 2024
  North East Database Day 2024 — Boston, MA
- **Check Out the Big Brain on BRAD: Simplifying Cloud Data Processing with Learned Automated Data Meshes** — September 15, 2023
  Cockroach Labs — Virtual
- **Learning-Based Creation of Data Mesh Architectures** — March 10, 2023
  North East Database Day 2023 — Boston, MA